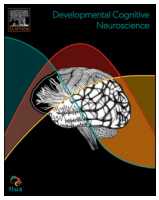
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The interplay between executive function and learning disabilities: Developmental cognitive neuroscience perspectives on reading, math, and ADHD

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ABSTRACT

The term "Executive function" (EF) describes domain-general cognitive processes including working memory, inhibitory control, and cognitive flexibility among others, that are essential for academic learning and behavioral regulation. Growing evidence highlights the importance of EF not only in typical development but also in understanding and intervening in learning disabilities (LDs), including dyslexia and dyscalculia, and in attention-deficit/hyperactivity disorder (ADHD). Despite this recognition, challenges remain in clearly defining EF, specifying its neurobiological underpinnings, and understanding its diverse roles across academic domains. This invited review reflects on a recent international expert workshop at the Flux Congress in 2024, and unites theoretical, behavioral, and neurobiological perspectives to examine how EF interacts with reading, math, and LDs. We explore the contribution of EF to word recognition, reading comprehension, and fluency; the role of audiovisual integration and cognitive control in dyslexia; the developmental interplay between EF and math difficulties; the role of EF in response to intervention in math; and EF as a transdiagnostic mechanism in ADHD and LD comorbidity. Across domains, we review the underlying brain networks, highlight methodological advances (e.g., functional connectivity, mobile neuroimaging), and discuss intervention strategies that target EF to support academic growth. We conclude with a synthesis of cross-cutting challenges and future directions, including the need for developmental models that integrate cognitive neuroscience, educational theory, and clinical insights. This review provides a roadmap for researchers and clinicians aiming to harness EF as a core mechanism to better understand and support diverse learners.

1. Introduction

Reading and mathematical skills are instrumental not just to academic success, but also to future career success and earning potential (Claessens and Engel, 2013; Meehan et al., 2023; Naveed et al., 2023). Academic skills form the foundation for how we communicate and trade information and skills with others, and thus academic skills predict general health and well-being (Hartog and Oosterbeek, 1998; Molla et al., 2004; Topitzes et al., 2009; Winkleby et al., 1992). Despite their high importance, many students struggle with academic skills; two-thirds of 4th graders in the United States are not proficient with reading, and more than one-third are below basic abilities in math, with many countries experiencing similar student difficulties (National Center for Education Statistics, 2023; OECD, 2023). Given their critical roles in modern society, it is unfortunate that many children experience learning difficulties (LDs) with math and/or reading that impact their

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future successes. Given the persistent nature of LDs, it is important to understand how they may interact with additional key cognitive processes, such as those underlying executive function (EF). The term EF encompasses that which is needed to achieve goal-directed, control-demanding behaviors. EF is slow to mature throughout development, activates the brain's control system, and directs a potentially malleable set of cognitive processes that could be harnessed to improve academic skills (Porter et al., 2023; Tervo-Clemmens et al., 2022).

Critically, however, there is no universal consensus on the definition of executive function (EF), and this can hamper fully understanding relations between EF and academic skills. As will be noted in more detail by Cirino, a variety of theoretical models have been proposed to conceptualize executive function (EF), and even within a given model, alternate solutions are found dependent upon age (Karr et al., 2018). These models provide distinct perspectives on the mechanisms underlying EF across different developmental stages and contexts. This lack of consensus on the definition of EF makes it challenging to create tasks that accurately or consistently isolate its components or inform relations to academics. Reading is a complex process that can benefit from a construct like EF as it requires the integration of orthography (understanding of printed language), phonology (understanding of language sounds and sound-to-print representations), and semantics (understanding of word meaning) (Coltheart et al., 2001; Seidenberg and McClelland, 1989). Decoding print to language also needs to be fast to enable fluent reading. As readers move from understanding individual words to passages of text, reading comprehension involves many more cognitively complex processes linked to EF (Gernsbacher, 1991). For example, holding the plot of a story in mind taxes working memory and other memory aspects to both understand the text and to link it to aspects of real life or similar stories of the past (Sweller, 2011).

Mathematic skills are also often considered to carry a high intrinsic cognitive load, involving the transition from controlled processes to more automatic mental math operations such as addition and multiplication (Sweller, 2011). Mathematic skills are built on cumulative knowledge and practice of facts, procedures, and complex applications (Van Hoof et al., 2017) and often require increased EF working memory capacity to support the learning of new math concepts (von Aster and Shalev, 2007). Although math and reading may seem quite different on the surface, they are strongly related, and difficulties in both often co-occur (Cirino, Child, et al., 2018; Peterson et al., 2017). Particular domains of math, such as word problems, also rely on reading comprehension (Fuchs et al., 2015). Moreover, several studies suggest that EF contributes to success in both areas (Carriedo et al., 2024; Cirino et al., 2015; Peterson et al., 2017). At the 2024 annual international Flux Society Congress, in Baltimore, Maryland, we (Horowitz-Kraus and Church) organized a pre-conference workshop to discuss the myriad of issues surrounding EF and academic skill research, particularly in those with LDs. We designed the workshop as both a primer for new researchers in the field, as well as a place to get top thinkers in the area to discuss their viewpoints regarding the role of EF in academic skill development with each other.

In this review, we have each workshop speaker providing a brief summary of their contributions to the workshop. We will provide an overview of the variety of EF definition(s), describe the brain activity related to EF, and the involvement of EF in aspects of reading and math brain activity and behavior, describe how EF impacts reading at the word-level and text-level in brain and behavior, and how auditory and visual integration of information is a key variable for reading.

We then move to mathematical abilities, and what we know about how EF intersects with math processing in brain and behavior, and how EF relates to math intervention outcomes. Our last section provides an important reminder of how LDs often co-occur with attention-deficit hyperactivity disorder (ADHD), and the possibility that the interrelation of these conditions occurs via EF. We finish with some closing thoughts about the future directions of EF work in the academic skill and LD research fields.

2. Conceptualizing executive function from a psychological perspective (Paul T. Cirino)

EF is wrought with confusion around how it is conceptualized and measured. The roots of EF trace back to unusual neurological case studies beginning in the 19th century and war injuries (typically involving the frontal lobe) in the early 20th century. In such cases, the dominant features were poor problem solving and decision making in everyday life, often in the context of preserved intelligence.

As the concept of EF evolved, two approaches predominated. One focused on individual differences in EF, generally focused on EF as a psychological construct (via objective or subjective measures). The other focused on EF as an anatomical or biological construct (via imaging or the study of patients with frontal-lobe injury). There has been, is, and will be, continued debate about the extent of complementarity of these two approaches (and their sub-approaches). Their connection is relevant in that, for example, functional imaging studies often examine activity during some task, the nature of which is informed (implicitly or explicitly) by the psychological lens through which one views EF. Given the size of these literatures, the present overview focuses only on the psychological approach, though of course an analogous review from a biological perspective would also be of benefit.

The cognitive framework for EF by Miyake and colleagues (Miyake et al., 2000), with its components of updating, shifting, and inhibition, is clearly the most widely utilized framework for measuring EF with performance measures (Karr et al., 2018). However, a number of other frameworks are available for how to construe what is important for achieving goal-directed behavior. Finding points of commonality and of differences among these is useful and relevant to help organize how to think about EF. Some alternative frameworks include the self-regulated learning process of Zimmerman (1990), the "macro-construct" of EF by Zelazo et al. (1997), the effortful control component of Rothbart's (2007) temperament model, and Engle's model of working memory (Engle, 2002, 2018). Cirino (2023) reviewed these with reference to their own prior definition of EF (Cirino, Ahmed, et al., 2018); that is, whether various frameworks considered EF to be: (a) singular versus plural, (b) domain specific versus general, (c) a process versus set of components; as well as (d) the extent to which EF implies control, or (e) management, or (f) a goal-directed target. That review found that there was widespread agreement across frameworks that EF was indeed domain general, and to some degree requires control, management, and goal-direction (although control and management were described in overlapping ways). However, there were also sharp distinctions, including whether EF should be construed as singular versus plural, and whether EF was a process versus a set of components. Cirino (Cirino, 2023) discussed the advantages and disadvantages of each of these distinctions, and argued for considering EF as a singular process, concluding that "*EF is a managerial process for coordinating domain general and domain specific resources during challenging goal-directed behavior*" (p. 10).

The process referred to in the above definition has three phases – those of representation, of planning, and of monitoring (the RPM model). Representation requires understanding that there is a task, problem, issue, or goal, and understanding its characteristics and context. Planning is developing an approach to go about reaching the anticipated goal state considering personal strengths and weaknesses, and monitoring refers to implementing the planned approach, including changing (or staying) the course, where needed. Such phases are not unlike those of other process models (Zelazo et al., 1997; Zimmerman and Schunk, 2011) and are also consistent with approaches taken to address "executive" difficulties (Levine et al., 2000; Ylvisaker and Feeney, 2009).

Given that the above approach is different from the component model of Miyake et al. (2000) and its derivations, it is necessary to resolve the way such components (shifting, inhibiting, updating) "fit". For example, there is robust evidence that each of those three constructs

are domain general and are related to a variety of goal-directed behaviors, that each is associated with certain brain regions and networks, and that there are common ways that each are measured. However, there is less attention paid to problems associated with this approach. For example, it is not clear how the components work together mechanistically, or how they "give rise" to EF per se. Further, it is not clear why other domain general cognitive factors (e.g., attention, processing speed) are *not* included. A componential approach also goes against the concept of EF as an individual difference, which is how EF is typically referenced. Finally, not all goal directed behaviors require the same combination of domain general (and domain specific) skills.

On the other hand, a major difficulty with a process approach to EF is how to measure it. To align with the proposed RPM model (Cirino, 2023), the phases should be measured directly. It may be possible to do so with a performance-based task, but this presents a number of issues, such as maintaining a balance between ultra-specificity and being overly abstract. However, developing such a task (or set of tasks) that have strong psychometric properties would be an excellent contribution. An alternative may be to develop a rating scale. Although multiple rating scales of EF are available (Barkley, 2012; Gioia et al., 2015), these do not approach EF from a process perspective, and in many cases, the measurement of EF overlaps significantly with that of behavioral attention symptomatology. Colleagues and I have (Cirino et al., 2026) developed a parent rating scale that directly instantiates the RPM model, with strong psychometric properties, good convergent validity, and better discriminant validity (relative to other EF scales) with regard to behavioral inattention. Although promising, this initial support needs to be evaluated in additional samples, via different raters (e.g., teachers), and against other measures and outcomes with which EF should be reasonably expected to relate (to varying extents).

There remains a need to resolve a number of outstanding issues with regard to EF. These include but are not limited to: (a) a continued reconsideration of how to conceptualize EF in a way that has strong theoretical and scientific yield but also is of practical relevance; and (b) how to measure EF with psychometrically robust instruments, particularly across potentially confounding approaches including material (e.g., verbal versus nonverbal), type (performance versus rating scale), and context (e.g., hot versus cool). Thinking carefully about what one "means by" EF (be it the RPM model of Cirino, 2023, a componential model (Miyake et al., 2000), or some other model), is an important first step for researchers (and clinicians) to consider before assessing or manipulating EF in a given context. Doing so systematically is likely to yield results for EF that are generalizable and replicable, and avoids a tendency for there to be a mismatch with an overly broad construal of what EF is, relative to how it is assessed.

3. Executive function and reading in the child brain (Jessica A. Church)

As noted by Cirino in the section above, many studies have investigated the engagement of brain systems related to EF. These investigations typically ask participants to perform a task designed to elicit control engagement while a neuroimaging device (electroencephalagraphy – EEG; functional near-infrared spectroscopy – fNIRS; functional magnetic resonance imaging – fMRI; magnetoencephalography – MEG) measures brain activity over time. Through comparisons of harder trials to easier trials, or trials that require an EF-related skill (e.g. "stop" trials relative to "go" trials to elicit inhibition in the stop-signal task (Verbruggen and Logan, 2008)) several brain regions have been shown to be consistently engaged and form putative "control networks" of the brain. Indeed, these EF-elicited brain regions correlate together even in the absence of a task, suggesting that these putative control networks have a history of working together over time.

However, also similar to Cirino's point above with the concept of EF, there is much debate and mixed vocabulary used to describe these EF-related brain regions, regarding whether they form a single or multiple brain networks, how overlapping they are across tasks, what cognitive processes are supported by these networks, and whether their relative engagement relates to behavioral EF (Engelhardt et al., 2019; Friedman and Robbins, 2022; Nugiel et al., 2020; Ribeiro et al., 2024). In general, it is agreed that controlled processing engages lateral frontal and parietal regions of the brain, along with the anterior cingulate and bilateral frontal operculum/anterior insula (Posner and Petersen, 1990; Power and Petersen, 2013). At rest, the lateral frontal and parietal regions are often reported to form one network (the "fronto-parietal" network), while the bilateral frontal operculum and anterior insula form another (the "cingulo-opercular" or "action mode" network) (Dosenbach et al., 2008, 2025; Power and Petersen, 2013). Critically, these brain networks are consistent with the lesion literature of patient symptoms, as well as with more direct recording data from surgical patients or non-human primates (Aron et al., 2004; Bunge et al., 2005; Gratton et al., 2018).

Work from my lab and others has found that these control networks are engaged during any circumstance when control is needed (e.g., when something is harder relative to easier), and that the engagement of these networks is consistent in location, if not in activity level, by at least middle childhood (Engelhardt et al., 2019; McKenna et al., 2017). EF-related behaviors (e.g., inhibition, working memory, cognitive flexibility) are slow to mature behaviorally, plateauing for most measured EF-related skills by 18 years or so (Porter et al., 2023; Tervo-Clemmens et al., 2022), and presumably there is continued maturation of control networks in the brain to support these behavioral effects over at least a similar time frame. What has been less clear is the role that these control brain networks play related to individual differences in academic skills and learning difficulties.

Similar to EF, reading and math tasks performed during neuroimaging have been shown to elicit consistent brain engagement. For reading tasks, brain activity is typically stronger in the left hemisphere relative to the right, and engages the ventral fusiform, middle temporal gyrus, supramarginal gyrus, and left inferior frontal gyrus among other regions (Martin et al., 2015; Schlaggar and Church, 2009; Schlaggar and McCandliss, 2007). Math tasks consistently elicit activity in the inferior parietal sulcus, along with many regions that are thought to relate to symbolic processing or language (Istomina and Arsalidou, 2024; Peters and De Smedt, 2018). Critically, for both math and reading tasks, even simple ones, the brain's control networks are heavily recruited, and often overlap with some of the strongest reading and math findings (e.g., inferior parietal cortex, lateral frontal cortex). Many of the researchers at this workshop have been fascinated by the possibility that understanding control network engagement during academic tasks may lend greater insight into those who struggle.

For example, in my lab we have found in upper elementary students that engagement of a parietal region of the fronto-parietal network, and the anterior cingulate region of the cingulo-opercular network during a sentence comprehension task related to reading performance outside of the scanner, but that engagement of these same regions during a non-lexical EF task (inhibition task), did not relate to reading (Roe et al., 2018). This result suggested engagement of control *during* a specific skill (i.e. during reading) may be altered in those with difficulty in that skill (i.e., struggling readers). However, a recent study of ours in bilingual middle schoolers complicates this summary, as differences in a non-lexical EF task (cognitive flexibility) *did* relate to reading ability (Nugiel et al., 2023). Furthermore, reading has also been shown to elicit unique multi-network integration properties, pulling control networks, the visual network, attention networks, and language regions together, and reducing network segregation, suggesting that children need multinetwork engagement for reading success (Church, 2023; Ellenblum et al., 2019; Lex et al., 2025; Vogel et al., 2013). While this section has focused on reading, in the workshop Wilkey noted exciting advancements in our understanding of how the brain's control networks support and relate to math abilities.

Overall, it is clear that EF, reading, and math can all be studied in the

brain through a variety of techniques and through testing a variety of different brain signals. How the brain's EF control networks dynamically correlate during different tasks or at rest, or how the brain engages control networks to perform a given academic task have already yielded unique insights about academic abilities. Hopefully more neuroimaging examinations of EF and math and reading in the future will allow us to better understand variations in academic achievement, better identify those at risk or why some struggle, and to develop targeted, effective future interventions.

4. Executive function and reading outcomes: neurobiological and cognitive linkages (Laurie E. Cutting; Sarah S. Hughes-Berheim)

Reading is a multifaceted cognitive process essential for academic success and lifelong learning. Reading relies on various cognitive processes, including decoding, vocabulary, and syntax, to integrate existing background knowledge with new information. The integration of these processes is hypothesized to be facilitated by EF. Findings from our ongoing lines of investigation, which we detail below, have illuminated the role of EF in reading outcomes, including revealing how impairments in EF can adversely affect reading abilities, as well as how enhanced EF may serve to facilitate outcomes for weaker readers.

While components of EF (i.e., the Miyake model (Miyake et al., 2000)) components of inhibition, working memory, and cognitive flexibility) have been linked to both word- and text-level processing (Spiegel et al., 2021), studies confirming *how* they are related to reading is an ongoing area of investigation. Previous hypotheses on the relationship between inhibition, working memory, and cognitive flexibility include that inhibition is important for enabling readers to attend to relevant word-level information (Coltheart et al., 2001; Henderson et al., 2013) or discourse while filtering out or suppressing irrelevant information or memories (Borella et al., 2010). The relevance of working memory during reading is that it enables readers to retain information about the text and integrate it into with their prior knowledge (Cain et al., 2004). The ability to continuously update information is a key component to learning from text (for review see Butterfuss and Kendeou, 2018). Current hypotheses regarding the importance of cognitive flexibility for reading includes it aiding readers in shifting between decoding words and retrieving semantic information from text (Cartwright et al., 2017; Taboada Barber et al., 2021) as well as preventing preservation during reading when presented with syntactic or semantic irregularities (Kieffer et al., 2013).

Recent studies from our lab (Hughes-Berheim et al., in preparation; Spencer et al., 2020) and others' (Cartwright et al., 2020; Kieffer et al., 2021; Nouwens et al., 2021; Taboada Barber et al., 2021) have highlighted EF appears to play a "behind the scenes" role, demonstrating EF specifically influences reading comprehension through word-level and oral language processes. Spencer et al. (2020) used a structural equation modeling (SEM) approach to determine that working memory and cognitive flexibility were indirectly related to reading comprehension via decoding and oral language, respectively, in a sample of 271 children ages 9–14 years old. Expanding upon this work, Hughes-Berheim and colleagues (in preparation) used a similar SEM approach to examine these relationships between EF (inhibition, working memory, and cognitive flexibility) decoding, oral language, and reading comprehension longitudinally (i.e., from grades 1–3; 2–4; and 3–5). Like Spencer et al. (2020), results showed that EF ability supports subsequent decoding and oral language, which further enhances later reading comprehension. Further, results showed a developmental shift with the relationship between EF and oral language increasing in importance for reading comprehension over time. Of note, these findings are compatible with the supposition that EF's role in reading (as well as other academic tasks) likely is a faciliatory one – helping to coordinate and enhance a diverse range of critical cognitive processes that need to be employed while reading.

In addition to behavioral findings, neuroimaging techniques such as functional magnetic resonance imaging (fMRI) have provided insights into the brain networks involved in reading and EF. Indeed, specific brain regions, including the bilateral dorsolateral prefrontal cortices (dlPFC), intraparietal sulci (IPS), and lateral medial temporal lobes, have been identified to support different aspects of EF; collectively, these brain regions form the fronto-parietal network (FPN) (Yeo et al., 2011) as noted earlier and in our recent work. Notably, specific regions within the FPN overlap with the reading network, suggesting that, at the neurobiological level, reading necessitates engagement of executive processes. Specifically, theories suggest the FPN is involved in strategically integrating newly read information with background knowledge to improve comprehension. This hypothesis is supported by research indicating the way reading/language regions interact with nodes within the FPN is linked to reading outcomes at both the word- and discourse-level (Aboud et al., 2018; Ihnen et al., 2015; Norton et al., 2014).

Word-level. Growth in word-level abilities have been linked to neurobiological indices of EF. In Aboud et al.'s 2018 paper, we set out to understand whether children identified with reading difficulties (i.e., Dyslexia) show unique functional activation patterns in regions associated with EF during reading tasks. Further, we explored whether these activation patterns were predictive of remediation following a tutoring intervention. Results indicated those with more efficient brain functioning in these areas (i.e., more connectivity between FPN and reading networks) were likely to demonstrate greater responsiveness to interventions. This sheds light on the potential for individualized intervention strategies that align with a child's unique neurobiological profile. In Burgess and Cutting (2023), we expanded upon this work and showed kindergarten EF was significantly associated with the right dorsolateral prefrontal cortex (dlPFC), part of the FPN, during a pre-reading fMRI task. Further, functional connectivity scores between the right dlPFC and the left ventral occipito-temporal cortex (vOTC; a region involved in word identification) predicted growth in reading comprehension ability a year later (Burgess and Cutting, 2023).

Together the findings of Aboud et al. (2018) and Burgess and Cutting (2023) suggest that neuroimaging in poor readers and/or pre-readers may at some point in the future enable identification of biomarkers of future reading difficulties. Specifically, they suggest early measures of EF function and network connectivity, may be used to identify students that may struggle to read long before children show signs of a reading difficulty. While the promise of using neuroimaging metrics in educational practice to determine what intervention approach may work for different neurocognitive profiles is a far-reaching goal, these and related studies provide a step forward towards this objective.

Text-level. Studies show that reading connected texts – or discourse processing – relies on the same brain regions that are active during word-level processing, while also encompassing broader reading regions and networks (Bailey et al., 2018). In particular, the default mode network (DMN), which encompasses the dorsomedial prefrontal cortex, bilateral angular gyri (AG), precuneus/posterior cingulate cortex, bilateral anterior temporal lobes, and hippocampus (Buckner et al., 2008), has often been found to play a role in discourse processing (Yang et al., 2019). The vast majority of discourse processing fMRI studies have used narrative texts, or stories (Keller et al., 2024; Yang et al., 2019), which intuitively align with the putative function of the DMN – historically, its involvement has been hypothesized due to theory of mind (Jacoby and Fedorenko, 2020; Saxe, 2006).

Nevertheless, the few studies of expository texts, including ours, suggest not only that expository text comprehension involves the FPN network, but also parts of the DMN (Aboud et al., 2019; Banich et al., 2023; Jacoby and Fedorenko, 2020; Ozernov-Palchik et al., 2021; Swett et al., 2013; van Moort et al., 2020; Wang et al., 2020). These results suggest certain areas of the DMN are integral to discourse processing, regardless of text type. Indeed, a recent conjunction analysis of ours (Hughes-Berheim et al., 2024), demonstrated that while the DMN is

more broadly involved in narrative text processing, a set of core overlapping areas within the DMN are present during expository text comprehension. Interestingly, degree of activity within these core discourse processing areas of the DMN was related to reading comprehension ability (Hughes-Berheim et al., 2024), suggesting these areas may be important for predicting the amount of material individuals learn from texts.

These findings emphasize the importance of large-scale networks, reading, FPN, and DMN that integrate functionally to complete the high-order task of comprehension. Of note, our recent findings align with contemporary hypotheses of the role of the DMN that propose it captures and represents individual narratives, including background knowledge and individual language experiences (Menon, 2023). During discourse comprehension, these experiences are drawn upon to create strong and robust mental models of the text (i.e., situation model building) to enhance comprehension.

4.1. Implications for educational practice

Understanding the relationship between EF and reading outcomes has implications for reading interventions and educational practices. Targeted efforts that simultaneously strengthen EF and reading skills may yield more significant improvements in literacy development than reading-only interventions. However, integration of EF training and reading intervention has been tried, and success on this front has been elusive thus far (Kassai et al., 2019; Melby-Lervåg et al., 2016; Takacs and Kassai, 2019). There recently has been interest in the field in thinking about EF in a more nuanced manner, specifically considering the context in which one needs to employ EF. Along these lines, people have discussed the idea of "reading specific EF" (Cartwright and Palian, 2024) and the analogous math specific EF (Wilkey, 2023a). A recent study demonstrated that reading-specific EF measures were superior to EF measures alone in predicting reading outcomes (Peng et al., 2024). Thus, it may be that "reading specific EF" tasks are needed to have any impact on reading outcomes.

This synthesis of our ongoing work highlights the critical relationship between EF and reading outcomes. Research demonstrates that enhanced EF is linked to improved reading proficiency; nevertheless, the mechanisms by which this occurs remain elusive. The impact of text genre on reading skills underscores the need for tailored interventions that address the varying demands of different texts. Future research should continue to explore the dynamic interplay between EF and reading, using both neural network and behavioral approaches, to identify effective intervention strategies that support struggling readers. By fostering both cognitive and reading skills, educators can help bridge the gap for students facing literacy challenges, ultimately promoting academic success.

5. Executive function and audiovisual integration (Tzipi Horowitz-Kraus)

The ability to read fluently (i.e., accurately and effortlessly with the appropriate prosody) is critical for reading comprehension (Breznitz, 2001). Difficulties with reading fluency, a somewhat under investigated subcomponent of reading, is often shared among children with dyslexia, but is also seen in other neurodevelopmental and neurological disorders (Breznitz, 2001). The reason for this shared fluency challenge is probably due to the complexity of synchronizing several sensory (audiovisual integration) and cognitive (EF) systems in the brain, which will be discussed in the following section (Breznitz, 2001).

As noted by Church above, reading is not a simple task but rather an interactive and integrative sensory and cognitive process involving the visual (also referred to as the orthographic processor), auditory (also referred to as the phonological processor), and semantic systems, described also in the Simple View of Reading model (Nevo and Breznitz, 2011; Scarborough, 2001). These systems are jointly engaged in the reading process with the visual system processing information slower than the auditory one (Breznitz, 2001; Hoover and Tunmer, 2020). Incoming visual information reaches the visual cortex about 70 ms post-stimulus onset, while auditory sensory information arrives in the auditory cortex about 30 ms following stimulus onset (Heil et al., 1999). On the other hand, viewed more broadly across the brain, linguistic information is proposed to be processed such that phonological processing is sequential (Rosenzweig and Bennett, 1996), whereas orthographic processing is more holistic (Breznitz and Leiken, 2001; Kraus and Breznitz, 2009), and thus output from the orthographic (more visual) processing systems may arrive faster than output from the phonological processing systems (Breznitz and Leiken, 2001). These two layers of potential timing mismatches lead to the importance of synchronization between the auditory and visual modalities for successful word reading in order to achieve fluency (Breznitz, 2001). It also points to the potential importance of higher-order processes, such as EF, in reading (Kim, 2020).

Individuals with dyslexia are believed to suffer primarily from a phonological processing deficit, contributing to inaccurate and/or slow reading (Shaywitz and Shaywitz, 2005). The definition and characteristics of dyslexia have undergone many changes over time, and newer theories also emphasize the role of EF in reading achievements among this population (Abo-elhija et al., 2022; Aboud et al., 2016; Horowitz-Kraus, Meri, et al., 2024; Lex et al., 2025; Nguyen et al., 2020; Spencer and Cutting, 2021). It is essential to distinguish between reading accuracy and reading fluency, with the latter being a combination of accuracy and speed (Breznitz, 2001; Perfetti, 1986; Snowling and Hulme, 1989). One aspect that affects fluency is speed of processing, which refers to the time it takes for information to be processed, including sensory and short-term working memory (Atkinson and Shiffrin, 2003). Speed of process and fluency are usually affected by word recognition efficiency, which is influenced by both orthographic and phonological processing (Breznitz and Misra, 2003).

Neurobiological studies, mainly using functional MRI, have pointed to altered involvement of the dorsal processing stream (i.e. phonological processing; including the supramarginal and angular gyrus) and ventral processing stream (orthographic processing, including the fusiform gyrus), as well as more frontal regions (related to the motor articulation and syntax of reading, and to semantic knowledge retrieval in the inferior frontal gyrus) in children with dyslexia (Shaywitz et al., 2003; Shaywitz and Shaywitz, 2008). Electrophysiological (EEG) studies have found that individuals with dyslexia may have more significant time gaps between auditory and visual recognition (Breznitz and Misra, 2003; Raij et al., 2000), leading to slower and less accurate processing, especially in recognizing imaginary words and complex formats (Breznitz and Misra, 2003). Dyslexic readers who were presented with words had significantly longer latencies in the P300 component, indicating delayed information processing and asynchrony between the visual and auditory modalities (Breznitz and Misra, 2003). The study concluded that these characteristics indicate difficulties in integrating orthographic and phonological representations, which are critical factors in word recognition in dyslexics (Breznitz and Misra, 2003).

A recent neuroimaging study demonstrated that during contextual reading, children with dyslexia show reduced functional connectivity between the auditory and the visual networks compared to age-matched 8–12-year-old typical readers (Horowitz-Kraus et al., 2023). However, a contextual reading fluency condition, which was achieved using a fast, rhythmic presentation of letters on the screen, resulted in increased functional connections between the auditory and visual networks in children with dyslexia, bringing their signals closer to those of typical readers. Interestingly, this increased functional connectivity was also related to increased reading comprehension in these readers (Horowitz-Kraus et al., 2023). These findings, together with morphometric studies pointing to the difference in gray matter for (word vs contextual) reading fluency in children with dyslexia (Horowitz-Kraus, Dudley, et al., 2024) emphasize the importance of audio-visual

integration during contextual reading for reading comprehension and raises the question regarding the role of EF in this integrative process among children with dyslexia, as suggested by Zhou et al. (2020).

Audio-visual integration (AVI) refers to the process by which the brain combines auditory (sound) and visual (sight) information to form a unified perception enabling the detection of speech in noisy settings, localizing sound sources, and processing synchronized stimuli like lip movements matching spoken words (Driver and Noesselt, 2008; Gao et al., 2023). This integration is crucial for interpreting and responding to complex sensory environments (Shams and Seitz, 2008). Efficient AVI improves how conflicting information from different senses is reconciled (Calvert and Lewis, 2004). Interestingly, speech-related audio-visual integration (AVI) and non-speech AVI demonstrate distinct neural mechanisms, as evidenced by developmental differences observed in previous studies. Speech-related AVI relies on specialized neural processing mechanisms that gradually develop with age and experience, while non-speech AVI is rooted in general sensory integration processes that mature early in childhood (Calvert and Lewis, 2004; Shams and Seitz, 2008). Several models have revealed that AVI is also related to interpretation and expectation processes, which are attributed to additional cognitive abilities, such as EF (Zhou et al., 2020).

The components involved in the AVI process occur in higher-level brain regions that process complex information. In this model, visual and auditory signals are initially processed separately in their primary sensory areas, the visual and auditory cortices, before being sent to more associative and integrative areas like the superior temporal cortex (STC) and prefrontal cortex (PFC) (responsible for decision-making and higher-order cognitive processing) (Gao et al., 2023; Stanford and Stein, 2007).

This model emphasizes the importance of top-down and bottom-up processes in AVI as well as the critical aspect of timing. Top-down processes, including cognitive processes, such as attention, expectations, predictability, and prior knowledge, play a critical role in shaping integration. For example, focusing attention on a particular stimulus can enhance its prominence, improving synchronization and perception (Talsma and Woldorff, 2005). Bottom-up processes are stimulus-driven characteristics, such as intensity, timing, and spatial proximity, that influence integration (Talsma and Woldorff, 2005). For instance, stimuli that are closely aligned in time and space are more likely to be perceived as connected (Dixon and Spitz, 1980).

Timing also plays a fundamental role in AVI, and is referred to in the AVI model as the time binding window (TBW) (Dixon and Spitz, 1980). TBW is the short time frame within which auditory and visual signals are perceived as synchronous. It allows the brain to merge signals arriving at slightly different times, enabling seamless interpretation of events. This TBW is not fixed; it can expand or contract based on factors like stimulus complexity, attention, and learning experiences. Predictable stimuli, like repeated speech rhythms, or positive emotions, such as happiness, expand the TBW, enabling smoother integration. The TBW changes across the lifespan and differs in individuals with neurodevelopmental and neurological disorders (Woynaroski et al., 2013).

Traditionally, the reading literature has focused effort on specific phonological processing interventions for struggling readers (Ahmed et al., 2020). However, my lab and others have recently taken a novel direction, where we focus on training AVI and EF, aiming to improve reading fluency (Cancer et al., 2022). These AVI-based reading interventions used either rhythm-based auditory (Cancer et al., 2020, 2022) or visually-based stimuli (Horowitz-Kraus et al., 2014a, 2014b; Taran et al., 2023) and were shown to improve reading fluency. The auditory-based rhythmic training investigated the benefits of incorporating visual and auditory cues in reading training for Italian children with dyslexia (Cancer et al., 2022). The study found significant improvements in reading fluency abilities under both auditory and visual rhythmic training conditions. Furthermore, the impact of AV synchronization was examined and showed that while visual cues contributed to engagement, the auditory component remained the primary driver of reading improvement (Cancer et al., 2022). The study findings indicate a significant improvement in reading speed both immediately following the intervention and after three months, suggesting that an auditory-rhythm-based training has lasting effects on reading development (Cancer et al., 2022).

In the visual domain, studies examining the effect of a visual-rhythmic training on reading fluency and related neural circuits have demonstrated a similar positive effect. A visual rhythmic reading intervention (VRRT) was achieved by deleting the letters from the screen while presenting text (Taran et al., 2023). The letters were deleted one-by-one from the screen, initially based on the child's own reading speed, but following several correct responses for the comprehension questions that followed each of the reading trials, the deletion rate increased in 5% increments (and slowed down in the case of erroneous responses). Fast deletion of these letters also engages a number of EF-related skills: the reader has to inhibit themselves from gazing backward (inhibition), they learn to process more letters in a given time (working memory), and speed of processing is potentially improved (Farah et al., 2024; Horowitz-Kraus, 2016; Horowitz-Kraus et al., 2014a, 2014b; Taran et al., 2023). Mechanistically, it was proposed that this manipulation releases the bottleneck of working memory in those with dyslexia and hence AVI occurs more efficiently with a change in strategy of more orthographic reading than decoding attempts (Breznitz and Share, 1992).

Training with this program improved reading fluency, accuracy, and comprehension in several studies conducted in children with dyslexia and matched typically-developing Hebrew (Horowitz-Kraus et al., 2014a, 2014b; Horowitz-Kraus and Breznitz, 2014) and English-speaking children (Horowitz-Kraus and Holland, 2015). Not only did reading measures improve, but also EF measures did as well (Horowitz-Kraus and Holland, 2015). Neurobiological measures have the advantage of identifying the neural circuits/pathways underlying changes in performance. After an 8-week training (3 times/week, 20–30 min per session), increased functional connectivity was observed between an EF-related network (cingulo-opercular) and the left fusiform gyrus (Taran et al., 2023), increased functional connectivity within the cingulo-opercular network (Horowitz-Kraus et al., 2015), and increased functional connections between the sensory networks (visual and auditory) and cingulo-opercular and fronto-parietal control networks, and all related to improved word reading fluency (Farah et al., 2024; Horowitz-Kraus et al., 2025). A moderation model demonstrated that changes in EF (specifically inhibition and speed of processing) were related to long-term reading fluency gains (Taran et al., 2022). Taken together, these studies support the role of AVI and EF in reading fluency among children with dyslexia as well as in typical readers and highlight the role of both the sensory system and EF in reading fluency.

In conclusion, reading fluency is a complex process relying on several sensory and cognitive processes, including AVI and EF. Targeting these processes hold promise, especially in populations suffering from reading fluency deficits, including dyslexia, but not limited to it. Individuals with low and high ADHD traits (Panagiotidi et al., 2017), those with Autism Spectrum Disorder (Solari et al., 2017), and individuals with epilepsy (Vlooswijk et al., 2010) all suffer from impairments in either AVI, EF, or both, which may contribute to their reading fluency abilities. Additional studies should characterize the similarities and differences in reading fluency and underly mechanisms in these populations and examine the effects of rhythmic interventions on their reading fluency skills.

6. Executive function and math learning difficulties: how we define EF in context and what that means for measurement (Eric D. Wilkey)

Individual differences in EF consistently relate to a range of academic skills (Arrington et al., 2014; Draper et al., 2025; Mazzocco and Kover, 2007; Willoughby et al., 2019; Draper et al., 2025). However, EF often

has a unique relation to mathematics learning based on the cognitive demands of specific math skills such as rational number development (Gilmore et al., 2024) and problem solving (Blair et al., 2008) for a meta-analysis, see Spiegel et al. (2021). Moreover, persistent and severe difficulties learning mathematics (developmental dyscalculia) often co-occur with EF challenges (Agostini et al., 2022). Because EF is "domain-general" (i.e., involved in a variety of goal-directed behaviors, sensory modalities, and stimulus content) we must ask how and why it relates specifically to math. Although knowledge of the neural bases of mathematical cognition and EF is advancing, a clearer account of their mechanisms, and how they interact, is essential for answering questions about the role of domain-general cognition in math learning difficulties.

Two recent neuroimaging meta-analyses provide some degree of consensus on the brain regions associated with math learning difficulties. Martinez-Lincoln et al. (2023) reviewed studies of math and reading difficulties and found over-activation of the right inferior frontal gyrus (IFG)(extending into the insula) and underactivation of the right anterior intraparietal sulcus (IPS) were most consistently associated with math difficulties. Tablante et al. (2023) identified these same regions and additionally reported underactivation of the ventral occipitotemporal cortex (i.e., fusiform and inferior temporal gyri) in the math difficulties group.

There is less consensus about which types of processing drive these neural differences. Tasks contributing to the IPS cluster included math-facts processing, magnitude and ordinality judgments, color comparison, spatial working memory, and transitive reasoning. Tasks contributing to the IFG cluster primarily involved ordinality and math facts. Moreover, while the ventral occipitotemporal cortex is known to have a more circumscribed role in symbol-based visual expertise, the frontal and parietal regions identified in these analyses have a wider range of functions. These functions do include magnitude processing (Sokolowski et al., 2017), but they are also frequently associated with attentional allocation (Connolly et al., 2016), visuospatial working memory (Klingberg, 2006; Silk et al., 2010), control of grasping and reaching (Grefkes and Fink, 2005), and task difficulty more generally (Bankó et al., 2011; Bokde et al., 2001; Gould et al., 2003). Therefore, although the two meta-analyses identify regions involved in mathematical cognition, regional specification at this level does not answer the fundamental question of which neurocognitive mechanisms drive processing differences in children with math-learning difficulties.

Addressing this question, the largest fMRI study of developmental dyscalculia to date tested three theoretical accounts of processing differences to determine which tasks elicited atypical brain activation (Kwok et al., 2023). All children (dyscalculic and control, 8–10 years old, n = 68) completed three tasks: (1) magnitude processing task, (2) arithmetic fluency, and (3) visuo-spatial working memory. Ultimately, Bayesian analyses suggested that there were no consistent differences in any of the tasks with the preregistered fMRI contrasts despite robust behavioral differences in the tasks.

The authors proposed that sample heterogeneity may account for the null findings. Group-level analyses, such as mean comparisons, assume that children with dyscalculia differ from typically developing peers in a consistent way across the sample. However, children with math learning difficulties are well-documented to be highly heterogeneous. This variability has led researchers to propose multiple subtypes of math difficulty, involving deficits in working memory, attention, spatial processing, and comorbid learning disorders (Bartelet et al., 2014; Chan and Wong, 2020; Fias et al., 2013; Landerl et al., 2013; Skagerlund and Träff, 2016; Szűcs, 2016). Given this heterogeneity, detecting mean differences in brain activation between groups may not be feasible.

Interpreting individual differences in EF adds another layer of complexity. If EFs are defined as higher-order processes that regulate lower-level cognition to achieve a goal (e.g., mathematical reasoning or problem solving), then three elements must be considered: (1) the higher-order EF mechanism, (2) the lower-level cognitive substrate, and (3) their neurobiological interaction. This complexity complicates how we interpret the meta-analytic neuroimaging findings discussed above. Differences in activation could reflect not only processing specific to the target task (e.g., magnitude comparison) but also the EF demands that task imposes, and the dynamic interaction between EF mechanisms and domain-specific cognition.

Several studies suggest that interactions between lower-level cognitive mechanisms and higher-order EFs warrant closer investigation. For example, in a large study of typically developing young adults (n = 546), participants completed multiple EF tasks in the scanner, including an antisaccade task (inhibition), a 'keep track' task (working memory), and a number-letter task (switching) (Reineberg et al., 2022). Common activation was observed across EF tasks, consistent with shared domain-general neural resources. However, correlations between brain activation and behavioral performance revealed a different pattern: individual differences in performance were associated not with activation in these domain-general EF regions, but with activation in task-specific target regions—the neural substrates upon which EF processes operated.

A similar pattern emerges in math-specific contexts. In one study, 3rd- and 4th-grade children completed a magnitude comparison task under two conditions: comparisons of dot quantities with matched visual stimuli and comparisons involving misleading visual cues (Wilkey and Price, 2019). In the interference condition, the less numerous set was more visually salient (e.g., larger surface area and greater average dot size) in half of the trials, increasing inhibitory demands. Only individual differences in this interference contrast—not the basic magnitude comparison—predicted math achievement. Importantly, this relation persisted after controlling for brain activity during the Eriksen Flanker task, an interference-suppression task in a non-numerical domain. In other words, it was magnitude processing under conditions of heightened EF demand, rather than magnitude processing alone, that explained variance in math performance.

Across these studies, the relation between brain function and behavior could only be interpreted by considering (1) the tasks' EF demands, (2) the lower-level cognitive processes involved, and (3) their neurobiological interaction. This highlights a critical recommendation: researchers must carefully consider which aspects of domain-specific processing, domain-general EF processes, and their interaction are being measured, intentionally manipulating each piece of these components (Wilkey, 2023b). Although EFs play a well-established role in the development of academic skills—particularly mathematics—current neuroimaging evidence offers an incomplete picture. Meta-analyses identify candidate neural substrates of atypical processing, yet the cognitive conditions under which these regions function differently remain unclear. Moreover, these conditions likely vary across individuals, given the heterogeneity within math-learning difficulties. Processing differences may reflect domain-specific mechanisms, EF mechanisms, or their interplay. Advancing this field will require refined methods for assessing EF and deliberate attention to the construct validity of tasks intended to measure EF in mathematical contexts.

7. Executive function in the context of mathematics interventions (Marcia A. Barnes)

The study of EF has grown rapidly over the past 40 years across many disciplines, including learning disabilities (Müller and Kerns, 2015). The relationship of EF to mathematics development and disability is robust (Peng et al. 2016), and stronger for mathematics compared to reading (e.g., (Morgan et al., 2019; Spiegel et al., 2021). However, because every child neurological, clinical, psychiatric, neurodevelopmental and learning disorder is associated with perturbations in EF, behavioral measures of EF are neither sensitive nor specific to the diagnosis of math disabilities. On the other hand, if EFs are affected by all childhood disorders perhaps they are important for cognition and learning, leading us to ask whether child EF is useful for understanding effects of interventions on learning and performance.

There are two analytic approaches to answering this broad question.

Each is associated with different types of questions that can be asked and answered. The first approach is moderation analysis (sometimes called *aptitude by treatment* interactions (Cronbach and Snow, 1977)), where we ask whether a child's level of EFs at the start of an intervention affects intervention response; for example, are intervention effects stronger for children with relatively lower levels of EF (a compensatory interaction effect), is the effect of the intervention stronger for children entering the intervention with relatively higher levels of EF (a synergistic interaction), or are treatment effects invariant across levels of EF (a universal effect of the intervention)? The second approach – mediation analysis- is one that allows inferences about causality and mechanisms of change. The question that is asked in mediation analysis is whether improvements in mathematics are mediated by improvements in EF. Evidence related to each approach in the domain of mathematics disabilities follows.

7.1. Does EF moderate response to intervention in children with or at risk for math disabilities?

Consistent with the relatively recent requirements to report moderator analyses in requests for applications from U.S. federal research agencies (e.g., Institute of Education Sciences), most of the papers reporting moderator effects in U.S.-based studies have been published in the last 10–12 years (Chow and Hampton, 2022; Spybrook et al., 2020). A systematic review conducted by our research group (Barnes, 2023) investigated whether child EF moderated effects of math interventions. Of the studies in the systematic review, the most frequently and consistently tested EF moderator is working memory. Math intervention studies testing working memory as a moderator are largely those addressing math word problem solving with both whole and rational numbers (e.g., (Fuchs et al., 2014, 2016; Swanson et al., 2013; Swanson, Moran, et al., 2014), consistent with the role of working memory in cognitive process models of math word problems (Kintsch and Greeno, 1985), and with meta-analytic findings that working memory is more strongly related to math word problem solving than to some other domains of mathematics (Peng et al., 2016). Most of these effects were synergistic; that is, children with relatively better developed working memory were more likely to gain more mathematical knowledge from the interventions. One study showed nuanced effects such that a fractions intervention with a practice component involving fractions computation fluency was most advantageous for students starting relatively higher in working memory, whereas the same intervention with conceptually based fractions practice (e.g., drawing and explaining fractions) was most advantageous for students starting relatively lower in working memory (Fuchs et al., 2016). Although cognitive strategy instruction is often part of effective intervention practices, Swanson and colleagues (Swanson, Orosco, et al., 2014) found that poorer math outcomes were associated with the attempted application of strategies during learning that exceeded the working memory capacity of students with math disabilities.

These findings suggest that one aspect of EF, namely working memory, is implicated in response to math interventions. These findings are important for answering the question of "for whom is this intervention most beneficial?". Barnes and Peng (2025) have argued that such findings are also relevant for telling us about the potential for hidden cognitive load in mathematics interventions. The preponderance of synergistic effects means that the content and procedures of several of our generally efficacious interventions produce extraneous load that taxes some children's cognitive capacity in ways that are not beneficial for learning (Sweller, 2010). Considering EF characteristics of both the learner (i.e., working memory level) and of the intervention (i.e., cognitive load requirements) has the potential to provide information that is important for engineering interventions to produce better learning outcomes for children with math difficulties and disabilities.

7.2. What is the nature of the relationship of EF and learning in children with or at risk for math disabilities?

Experimental studies (i.e. intervention studies) provide the level of evidence necessary to determine whether changes in mathematical learning are causally related to growth in EF. There are several meta-analyses of working memory interventions showing a lack of "far transfer" effects on mathematics (e.g., (Melby-Lervåg et al., 2016; Sala and Gobet, 2020), suggesting that increases in working memory do not lead to changes in mathematics learning or performance. The relevance of these meta-analyses for making inferences about students with math disabilities have been critiqued on several grounds (Fuchs et al., 2022). First, few studies had participants with math disabilities. Whether working memory training affects growth in mathematics in such individuals is unknown, but typically, interventions have larger effects for those who require intervention. Second, measures of math achievement in these studies are narrow in that they focus on arithmetic to the exclusion of other domains of mathematics that are strongly associated with working memory such as math word problem solving (Peng et al., 2016). Finally, very few studies test whether combined EF and math interventions differentially affect math outcomes compared to math-only or EF-only interventions (for a review of these types of studies see Barnes and Peng, 2025).

One study has tested the nature of the relationship of working memory and mathematics in the context of a randomized controlled trial (Fuchs et al., 2022). Second grade children with or at risk for math disabilities received either: a working memory intervention; a schema-based math word problem solving intervention; or the schema-based math word problem solving intervention with 6 min of embedded working memory games using math stimuli. Each of these three active treatment conditions was followed by a 5-minute math calculations and word problem solving practice, which may have provided an explicit link between working memory and mathematics for the working memory intervention group. These groups were compared to each other and to a business-as-usual condition. The working memory intervention showed strong effects on working memory and significant, but smaller effects on measures of math word problem solving. The two schema-based math interventions produced large effects on math word problems and significant, but smaller effects on working memory. These findings show that the relationship between working memory and mathematics may be bidirectional (Doebel, 2020; Peng and Kievit, 2020) because a working memory intervention produced positive changes in mathematics, and a mathematics intervention produced positive changes in working memory.

Mediation analysis, which tests purported mechanisms of treatment effects showed that: 1) the working memory intervention with a small amount of math practice led to growth in working memory, which, in turn, was related to growth in math word problem solving; and 2) the math interventions led to growth in math word problem solving both directly and also indirectly through (or mediated by) growth in working memory.

In conclusion, studying EF in the context of mathematics intervention reveals several findings of importance. First, moderation analyses show that a child's level of working memory often affects their response to mathematics intervention. Second, the *pattern* of moderation effects, namely that interventions were often more effective for children starting somewhat higher in working memory, highlights that there may be extraneous cognitive load in some of our generally effective math interventions. Third, the relationship of working memory and math word problem solving is bidirectional in that improving working memory improves math word problem solving, and vice versa. Finally, mediation analyses support the idea that the mechanisms behind intervention-related growth in math word problem solving involve both domain specific learning of mathematical content and procedures as well as changes in working memory. These findings are of importance to models of the development of math and EF and serve to underline for

researchers and practitioners alike, that development of EF may be malleable through engagement with complex academic tasks in children with math learning disabilities.

8. Executive function deficits as a transdiagnostic feature of ADHD and learning disorders: implications for assessment and intervention (Keri S. Rosch)

Attention-deficit/hyperactivity disorder (ADHD) is the most common neurodevelopmental disorder, with recent prevalence estimates of 10% among school-aged children in the U.S. (Danielson et al., 2022). The defining symptoms of ADHD include developmentally inappropriate levels of inattention, hyperactivity, and impulsivity that are present across settings and associated with functional impairment in academic, peer, or family domains (American Psychiatric Association, 2013). Individuals with ADHD often display deficits in EF, whether defined behaviorally (e.g., parent reports difficulty following multi-step directions) or measured with a neuropsychological test (e.g., ability to hold and manipulate verbal information in working memory). Academic difficulties are common in individuals with ADHD, often driving the referral for assessment or intervention, with concerns including poor grades, failure to complete schoolwork, difficulty retaining information, inconsistent academic skill demonstration, taking longer than expected to complete work, and feeling overwhelmed by schoolwork. Children with ADHD are more likely to obtain lower school grades and standardized test scores and higher rates of educational placement and grade retention (DuPaul and Stoner, 2003). In adolescence, ADHD is associated with higher rates of school drop-out and lower rates of college enrollment, which may contribute to lower vocational attainment and less job/financial stability in adulthood (Barkley et al., 2008; Mannuzza, 1993). Improving academic functioning in ADHD is critical to disrupting the negative, long-term trajectories that are characteristic of the disorder.

The diagnosis of ADHD is based on the behavioral symptoms listed above and does not include a brain scan or cognitive testing, but the symptoms refer to cognitive functions that affect academic performance (particularly the inattentive symptoms). For example, the symptom "often fails to give close attention to details or makes careless mistakes in schoolwork" may be expressed as comprehension errors while reading, spelling/grammar errors while writing, or errors while completing multi-step math problems. Also, the symptom "often avoids, dislikes, or is reluctant to engage in tasks that require sustained mental effort" may apply to schoolwork, resulting in less practice with academic skills, thereby exacerbating weaknesses in these areas for children with ADHD. ADHD frequently co-occurs with learning disorders, with up to 50% of children with one diagnosis meeting the criteria for the other (DuPaul et al., 2013). Given the high rates of comorbidity between ADHD and learning disorders, with poorer outcomes when both disorders are present, there is a need to better understand the mechanisms contributing to this comorbidity and novel interventions to improve outcomes.

The co-occurrence of learning disorders and ADHD is thought to be due, in part, to atypical development of shared neurobiological pathways supporting EF (Denckla, 1989; McGrath et al., 2011). ADHD is a heterogeneous disorder, both in terms of the symptom presentation, associated cognitive/motivational deficits, and clinical course. Multi-process models of ADHD postulate competition between neurobiological circuits involved in executive functions and reward response are thought to underlie the behavioral expression of ADHD symptoms (Sonuga-Barke, 2002, 2005). Among children diagnosed with ADHD, there is significant variability in the extent to which they show cognitive or motivational deficits, or atypical neurobiology. Perhaps for some children with ADHD, the executive circuit is primarily affected which is also implicated in LD, whereas the atypical motivation may be specific to ADHD.

Multiple deficit models (MDM) have also been applied to understand the etiology of learning disorders and their comorbidity with ADHD. First proposed by Pennington, MDM is a multi-level framework for understanding neurodevelopmental disorders, spanning etiology (genes, environments, and gene-environment interplay), brain mechanisms, neuropsychology, and behavioral symptoms. This model, first applied to understanding developmental disorders by Pennington (2006), accounts for inter-individual variability in profiles by proposing that each risk factor is probabilistically (rather than deterministically) related to neurodevelopmental disorders. Most research has occurred at the neuropsychological level of analysis. Studies have shown consistent support for the fundamental tenets of the MDM, that multiple probabilistic risk factors are associated with neurodevelopmental disorders and that shared risk factors contribute to comorbidity (McGrath and Stoodley, 2019). The MDM has been tested in a longitudinal twin sample which aimed to identify shared and unique cognitive correlates of learning difficulties and ADHD. Results of this and other studies from the Colorado Learning Disabilities Research Center group suggest that the shared variance of EF and processing speed explain most of the transdiagnostic risk for ADHD and LD (McGrath et al., 2011, 2020).

Following the introduction and research on MDM, researchers believed the field could eventually identify performance patterns across neurocognitive domains that would be diagnostic. However, the dimensional and probabilistic nature of the deficits prevents clear mappings of cognitive profiles to diagnoses. The heterogeneity of neuropsychological deficits among individuals with the same diagnosis (ADHD or LD) is now well-established, with some individuals showing only a single deficit or even no identifiable deficit. Therefore, diagnosis continues to be based on evaluation of the defining symptoms of the disorder (based on clinical history, observations, and reliable, validated tests) as well as careful attention to functional impairment and possible comorbid conditions. While detailed neurocognitive evaluation may not be necessary for diagnostic decision-making, cognitive profiles might still be helpful for treatment planning and for helping families and educators adapt to the unique cognitive strengths and weaknesses of the child (e.g., accommodations for working memory or processing speed challenges).

Given this background, there are several important assessment considerations when there are concerns for ADHD and learning. Among students with learning difficulties, parent and teacher ratings of ADHD symptoms is recommended to determine whether further evaluation of ADHD is necessary (e.g., if symptom ratings exceed a specific clinical threshold). In a similar fashion, classroom performance data (e.g., curriculum-based measurement probes) for reading, math, and writing can be used to determine whether further evaluation of skills deficits is necessary for students with ADHD. Assessment of academic functioning can help identify specific targets for intervention beyond reduction of ADHD symptomatic behaviors. Possible changes in academic functioning associated with treatment, including medication and behavioral interventions, should be assessed on a regular basis.

It is also important to consider the types of learning/academic difficulties the child is having and the level at which ADHD symptoms may be contributing to these difficulties. ADHD is associated with weaknesses in multiple levels of each academic domain, raising important questions that may be teased apart with neuropsychological assessment. For example, it is important to understand whether symptoms of ADHD are indirectly influencing advanced skills by undermining the development of foundational skills, or whether ADHD has a direct and unique impact on advanced academic skills, including reading comprehension, writing composition, and mathematical problem-solving. A recent study by Trane and Willcutt (2023) on the Colorado twin sample described above suggests a direct effect of ADHD on advanced academic skills, such that it independently predicted writing composition and reading comprehension after controlling for basic skills. However, individualized assessment is critical to guiding intervention approaches given the heterogeneity in ADHD and associated learning difficulties.

When thinking about intervention, the assessment should help us to

understand whether academic functioning is affected by academic skill-deficits, which are LD-related, or performance-deficits, which are ADHD-related. Performance deficits can presumably be addressed by behavioral interventions and/or psychotropic medication as alleviating ADHD symptoms may lead to better academic performance. Academic skill deficits must be addressed directly through empirically supported instruction in the relevant skill area. There are evidence-based behavioral interventions for youth with ADHD that target EF skills impacting academic performance (DuPaul et al., 2012; DuPaul and Eckert, 1997). These approaches focus on application of EFs to daily activities by teaching behavioral strategies. A behavioral intervention plan is also recommended, which emphasizes the use of clearly defined behavioral targets and expectations followed by frequent feedback and rewards along with facilitating communication between parents and teachers (Iznardo et al., 2020). There has also been a focus on cognitive training interventions in the ADHD literature, designed to directly target and improve EF and processing speed. These include brain training programs such as CogMed, Endeavor Rx, Interactive Metronome, and neurofeedback (Chacko et al., 2014; Enriquez-Geppert et al., 2019; Kollins et al., 2021; Park and Choi, 2017). These are appealing alternatives to the behavioral interventions described above as they are more easily implemented by families, but only seem to show near-transfer, meaning improved performance on that specific skill, but not far-transfer to improving ADHD symptoms or learning difficulties. Further research with randomized controlled trial methodology is needed to determine their effectiveness.

There is also value in using intervention as experimentation to clarify the role of EF learning, academic performance, and response to intervention. For example, we have previously demonstrated that, among children with ADHD, medication effects on classroom performance (math productivity and classroom behavior) were mediated by medication effects on certain cognitive functions (inhibition and working memory) (Hawk et al., 2018). This might suggest that children struggling to complete their classwork who have difficulties with inhibition or working memory may benefit from stimulant medication. Improved understanding of individual cognitive profiles that relate to response to intervention will allow for personalized, targeted interventions and inform our mechanistic understanding of learning/academic difficulties and effective interventions.

There is also value in using interventions with EF skill training embedded in the academic skill practice, such as the rhythmic reading training discussed earlier by Horowitz-Kraus, to probe the role of EF in academic performance. We recently published a study (Rosch et al., 2025) of children with reading difficulties with and without ADHD who completed the rhythmic reading training and examined reading and EF skills following the intervention. Results suggest the greatest improvement in reading comprehension was among children with comorbid ADHD and RD following the intervention. While children with reading difficulties with and without ADHD showed improvement in inhibition, which was normalized relative to the typical readers group, children with reading difficulties and comorbid ADHD showed greater improvement in reading comprehension. We also saw significant change in brain functional connectivity of cognitive control networks (fronto--parietal and dorsal attention) in the comorbid group. This is an example of how intervention can be used as experimentation, to test a theory of change or underlying cognitive mechanisms.

In conclusion, ADHD and LD frequently co-occur due in part to shared EF and processing speed deficits. Assessment of ADHD and LD should include screening of the other disorder. Intervention across home and school involving behavioral/medication for ADHD and explicit skill instruction for LD is recommended. Finally, development of interventions targeting shared cognitive deficits or individual cognitive profiles can inform theory and practice.

9. Discussion

9.1. Integrative synthesis and panel insights

The workshop held at Flux 2024 in Baltimore, Maryland, brought together a number of exciting and insightful perspectives regarding the interplay between EF, academic skills, and LDs in brain and behavior. Speakers covered a variety of overlapping issues, including issues of how EF is defined and measured (Cirino, Church, Rosch), how EF impacts reading (Church, Horowitz-Kraus, Cutting) and math (Church, Wilkey, Barnes) both behaviorally and with regard to intervention (Barnes, Cirino, Horowitz-Kraus), as well as neurally (Church, Cutting, Wilkey), and reflections on how EF impacts comorbidities with reading and math difficulties like ADHD (Rosch).

The consistent theme across the workshop was that we need a better understanding of how all the approaches to defining and studying EF fit together. Some research indicates that there is a common, cross-domain set of EF processes in the brain that are important to consider when studying LD, reading, math, or ADHD. EF, whether assessed by behavioral task measures, by self/parent/teacher surveys, or by brain measures, strongly relates to variation in reading and math abilities. However, the mechanisms by how EF impacts those with LDs and ADHD, and how EF impacts intervention or treatment response remains unclear. It is important that future work consider domain-specific and domain-general EF processes and interactions when trying to understand differences in academic performance across individuals and change in reading or math performance over time. Similarities of EF impact have been observed across reading and math scores, and across those with ADHD and/or LDs, but there have been important observed differences too, as far as the duration, consistency, and specificity of EF effects in a given study.

Critically, it remains unclear whether and how including EF aspects in educational interventions is helpful, and under what circumstances. In particular, the context in which EF is incorporated (within the academic context or more generally) may affect the duration and impact of EF training. All authors note the need for future work in this area to answer a number of enduring questions about the role of EF in learning interventions.

The end of the workshop held a panel with NIH program officers to discuss how EF work in the academic arena could have impact on policy and future research. The work clarifying EF's role in academic success and individual variation in ability is just beginning, and many exciting directions lie ahead.

9.2. Future directions

One thing we learned is that a variety of different neural approaches are used, and the overlap/consistency across them is unclear. A commonality is that EF has been linked to multiple putative control networks in the brain, but especially the fronto-parietal control brain network. As future brain studies work more towards network-level analysis, it will be fascinating to understand how EF abilities are supported by brain networks and to discover which networks show the most plasticity over age and learning change. How EF impacts reading and math processing in the brain may relate to how networks reconfigure, integrate, or segregate during different tasks.

Another thing we learned is that while it is clear that EF (regardless of approach) relates consistently to concurrent reading and math skills, it is unclear how to use that relation to improve educational outcomes over time or intervention. However, as the exciting work described by Barnes shows, EF and academic interventions may be bidirectional, and EF abilities may impact who benefits most from intervention. What is needed now are more studies that push the envelope on how we test those effects.

Relatedly, another set of future directions that remains critical with regard to the role of EF in academic success is to understand the impact

of subclinical levels of difficulty and the role of comorbidities on the individual. While LDs are often defined based on an individual scoring below a certain percentage of their peers, individuals just above that threshold are likely experiencing similar brain and behavioral effects. Similarly, ADHD symptoms form a spectrum, and the availability of assessment, diagnosis, and treatment is often not equally accessible across all strata of society. The high co-occurrence of ADHD with LDs, of multiple LDs (math and reading) with each other, and of LDs with anxiety and other mental health symptoms remain important directions for future EF research.

Large, longitudinal datasets that collect multiple measures of brain and academic skills are critical for clarifying the role of EF in LDs. We need to follow students over time to understand whether and how EF relates to academic skill change over development and within the same individuals. Most studies of academic skills in youth remain cross-sectional or, if longitudinal, are small, reflecting their inherent collection difficulty. Collaborative efforts that share and pool neuroimaging and behavioral collections in students across locations and over time could greatly advance our understanding not just of the relationships between EF and academics, but whether and how EF relates to academic skill change over development and within the same individuals. Studying brain and behavioral data across the academic calendar could greatly help us understand the nature of skill change in students at different stages of development. Data sharing will be instrumental in amassing reliable and robust effects.

To that end, increased use of more mobile neuroimaging technologies like fNIRS (Zhan et al., 2024), or portable EEG (Poulsen et al., 2017) are key technologies for future EF and academic research. They are quiet and cap-based, allowing more freedom of position. Low-field MRI is also an exciting developing technology that could allow more portable MRI collections (Kimberly et al., 2023). Being able to go out of the lab and into the classroom will remove biases inherent in lab-based collections, and allow studies of multiple students at once, or of moments of synchronization between teacher and learner.

As the speakers of the workshop make clear, current studies of EF, reading, and math are highly relevant to educators, parents, and students alike. Understanding more about EF and academics could allow for better and earlier identification of students at risk for future difficulty, could lead to more effective interventions by incorporating EF instruction in an evidence-based manner, and could suggest ways to tailor treatment to the individual. Much more work is needed, but many exciting research and educational directions lie ahead.

10. Conclusion

Understanding the intersection between EF, LD, ADHD, and academic abilities is a complex topic that needs interdisciplinary and developmentally sensitive approaches that reach across brain, behavior, and culture. It is somewhat early days for cognitive neuroscience studies of EF and math or EF and reading. We hope this summary, which emerged from the EF workshop at Flux 2024, is helpful for spurring discussion and guiding future research.

CRediT authorship contribution statement

Rosch Keri: Writing – review & editing, Writing – original draft, Methodology, Investigation. **Barnes Marcia:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Church Jessica:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization. **Cutting Laurie:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Cirino Paul:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Wilkey Eric:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Sarah S. Hughes-Berheim:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Tzipi Horowitz-Kraus:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare no conflict of interest.

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Data availability

No data was used for the research described in the article.

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